

Instagram + Facebook Bot Setup Guide

Student-friendly Thanglish step-by-step process

Goal	Telegram bot-la image create panni, Instagram/Facebook select panni, confirm kudutha post publish aaganum.
Project	D:\Ai_Agent\telegram-ai-bot\backend
Important	Indha project safety-kaga Preview -> Confirm Publish mudinja piraguthaan real post podum.

4 main values

Instagram token + Instagram account ID + public HTTPS URL + project encryption key.

Full flow - one line-la

```
Instagram Professional Account
|
Meta Developer App -> Token + Instagram ID
|
Backend + Cloudflare public URL
|
Telegram bot -> Generate -> Select -> Confirm -> Publish
```

Part 1 - Start panna munnaadi

1 Accounts ready pannunga

Oru Facebook personal login, oru Facebook Page, matrum oru Instagram account venum. Instagram account-ai **Business** allathu **Creator** account-a maathunga.

Instagram app-la:

- [OK] Profile -> Menu -> Settings and activity
- [OK] Account type and tools -> Switch to professional account
- [OK] Business allathu Creator select pannunga
- [OK] Mudinja varaikkum Facebook Page-oda link pannunga

Remember

Personal Instagram account-ai official publishing API use panni post panna mudiyathu. Professional account required.

2 Meta Developer App create pannunga

Browser-la **developers.facebook.com/apps** open pannunga. Login -> My Apps -> Create App. Business-related use case/app type select panni app create pannunga.

App create aanadhum:

- [OK] Instagram product/use case add pannunga
- [OK] Instagram -> API Setup with Instagram Login open pannunga
- [OK] Unga Instagram Professional account-ai connect pannunga

Dashboard labels

Meta dashboard button names konjam maaralaam. Aana thevaiyana section: **Instagram API with Instagram Login**.

Part 2 - Instagram keys edukkardhu

3 INSTAGRAM_ACCESS_TOKEN generate pannunga

Meta app-oda Instagram API Setup page-la **Generate Access Token** click pannunga. Instagram login/authorization complete pannunga.

Minimum publishing permissions:

```
instagram_business_basic
instagram_business_content_publish
```

Generate aana long secret value-ai copy panni .env-la podunga:

```
INSTAGRAM_ACCESS_TOKEN=IGAAxxxxxxxxxxxxxxxxxx
```

Secret

Indha token-ai screenshot, GitHub, WhatsApp group, email allathu public chat-la share pannatheenga.

4 INSTAGRAM_BUSINESS_ACCOUNT_ID edunga

Idhu Instagram username illa. Idhu Instagram API use pannura numeric user/account ID.

Token kidaicha piragu indha URL-ai Postman/browser-la use pannalaam:

```
https://graph.instagram.com/v21.0/me
?fields=id,username
&access_token=YOUR_ACCESS_TOKEN
```

Response example:

```
{
  "id": "17841400000000000",
  "username": "my_business_name"
}
```

Response-la irukkura **id** value-ai .env-la podunga:

```
INSTAGRAM_BUSINESS_ACCOUNT_ID=17841400000000000
```

ID confusion

Instagram Account ID enbadhu username, Facebook Page ID, Meta App ID kidaiyaadhu. Correct value usually 178... pola long numeric ID-a irukkum.

Part 3 - Public image URL setup

5 Backend start pannunga

Meta server unga generated image-ai internet-la download panna vendum. First local backend port 8000-la run aaganum.

```
cd D:\Ai_Agent\telegram-ai-bot\backend  
.\scripts\run.ps1
```

6 Cloudflare Tunnel start pannunga

Pudhu PowerShell window open panni indha command run pannunga. Indha window close panna koodathu.

```
cloudflared tunnel --url http://127.0.0.1:8000
```

Output-la ippadi oru public HTTPS URL varum:

```
https://happy-tree-example.trycloudflare.com
```

Adhai .env-la podunga:

```
SOCIAL_PUBLIC_BASE_URL=https://happy-tree-example.trycloudflare.com
```

Very important **localhost**, 127.0.0.1, http URL work aagathu. **https** public URL mattum use pannunga. URL-kku quotes/backticks thevai illa.

Quick Tunnel cloudflared restart panna trycloudflare URL maaralaam. Appo .env public URL update panni backend-ai restart pannunga.

Part 4 - SOCIAL_TOKEN_KEY

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Project encryption key generate pannunga

Indha key Meta-kitta irundhu varaadhu. Project script automatic-a generate panni .env-la safe-a ezhuthum.

```
cd D:\Ai_Agent\telegram-ai-bot\backend
python scripts/generate_social_key.py --write-env
```

Success output:

```
SOCIAL_TOKEN_KEY written privately to .env
```

Validate panna:

```
python scripts/generate_social_key.py --check
```

Expected:

```
SOCIAL_TOKEN_KEY is valid
```

Why needed?

Facebook OAuth token madhiri sensitive connections database-la encrypt panni save panna SOCIAL_TOKEN_KEY use aagum.

Do not change

Later indha key-ai random-a maathatheenga. Change pannina already saved encrypted connections read panna mudiyama pogalaam.

Ippo 4 main values ready

Variable	Enga irundhu?
INSTAGRAM_ACCESS_TOKEN	Meta -> Instagram API Setup -> Generate Access Token
INSTAGRAM_BUSINESS_ACCOUNT_ID	Graph API /me response-la id
SOCIAL_PUBLIC_BASE_URL	cloudflared kudukkura https URL
SOCIAL_TOKEN_KEY	Project generate_social_key.py script

Part 5 - Facebook connect pannardhu

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Facebook App ID + Secret edunga

Meta Developer App -> **App Settings** -> **Basic** open pannunga. Anga irukkura App ID matrum App Secret-ai copy pannunga.

```
FACEBOOK_APP_ID=your_meta_app_id
FACEBOOK_APP_SECRET=your_meta_app_secret
```

Secret

App Secret-ai public-a share pannatheenga. Idhu password madhiri.

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Facebook callback URL register pannunga

Meta app-oda Facebook Login settings-la indha exact Valid OAuth Redirect URI add pannunga.

```
https://YOUR-PUBLIC-HOST/auth/facebook/callback
```

Example:

```
https://happy-tree-example.trycloudflare.com/auth/facebook/callback
```

Bot request panna vendiya permissions:

```
pages_show_list
pages_read_engagement
pages_manage_posts
instagram_basic
instagram_content_publish
```

Development mode

App Admin/Developer/Tester accounts mattum testing-la work aagalaam. Public users-kku Live mode, App Review, Advanced Access allathu Business Verification thevai padalaam.

Telegram-la Facebook connect:

```
/connect facebook
```

Bot kudukkura link open -> Facebook login -> Page permissions allow -> callback success.

Part 6 - Final .env + testing

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Final values fill panni restart pannunga

backend/.env file-la placeholders illama real values podunga. Example keezha irukku; example values-ai copy pannatheenga.

```

INSTAGRAM_LOGIN_FLOW=instagram
INSTAGRAM_ACCESS_TOKEN=IGAA_your_real_token
INSTAGRAM_BUSINESS_ACCOUNT_ID=1784_your_real_id

SOCIAL_PUBLIC_BASE_URL=https://your-host.trycloudflare.com
SOCIAL_TOKEN_KEY=generated_key_already_written

FACEBOOK_APP_ID=your_real_app_id
FACEBOOK_APP_SECRET=your_real_app_secret

```

Backend restart panna piragu checks:

```

cd D:\Ai_Agent\telegram-ai-bot\backend
python scripts/check_instagram.py
python scripts/check_social_oauth.py

```

Telegram test flow:

1. Bot-kku /status anuppunga.
2. Facebook connect illana /connect facebook anuppunga.
3. Example: /image coffee shop offer poster anuppunga.
4. Generated image vandhadhum Instagram allathu Facebook select pannunga.
5. Preview correct-a irundha Confirm Publish click pannunga.
6. Success message/permalink vandhucha check pannunga.

Publish rule

Platform select pannadhu mattum post illa. **Confirm Publish** mudinja piraguthaan real social post create aagum.

Troubleshooting - common problems

Problem	Enna check pannanum?
Instagram token/account missing	Token and numeric account ID rendu values-um .env-la irukka check pannunga. Backend restart pannunga.
Instagram authentication failed	Token expire/revoke aagirukkalaam; correct Instagram Login token and correct account ID use pannunga.
Public media URL error	cloudflared run aagudha, current https URL .env-la same-a irukka, backend port 8000-la run aagudha check pannunga.
Facebook callback mismatch	Meta dashboard redirect URI exact-a SOCIAL_PUBLIC_BASE_URL + /auth/facebook/callback match aaganum.
No managed Facebook Pages	Login pannura Facebook user-kku Page full control/content permission irukka check pannunga.
Permissions error	Correct scopes approve pannunga. Public users-na App Review/Advanced Access thevai padalaam.
Restart-ku apram connection poiduthu	SOCIAL_TOKEN_KEY valid-a irukka; key-ai maathala; database path writable-a irukka check pannunga.

Final checklist

- [OK] Instagram Professional account ready
- [OK] Meta Developer app ready
- [OK] Instagram token + numeric account ID filled
- [OK] Backend and cloudflared running
- [OK] SOCIAL_PUBLIC_BASE_URL current-a irukku
- [OK] SOCIAL_TOKEN_KEY generated and valid
- [OK] Facebook App ID/Secret + callback configured
- [OK] Bot-la preview -> confirm test success

Official references

Meta Apps: <https://developers.facebook.com/apps/>

Meta Instagram API collection: <https://www.postman.com/meta/instagram/documentation/6yqw8pt/instagram-api>

Project README: D:\Ai_Agent\telegram-ai-bot\README.md